
Improving information gain for robustness in node-based Bayesian neural networks

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Abstract

Bayesian neural networks (BNNs) have emerged as a promising framework for uncertainty quantification and robust predictions under covariate shift. Recent advances in node-based BNNs simulate implicit input corruptions via latent node variables, with entropy regularization encouraging a diverse range of perturbations. However, simply maximizing entropy can yield uninformative latent samples and lacks a formal connection to output robustness. In this paper, we propose an augmented loss function that incorporates an explicit information-theoretic term. By modeling each latent variable as a bandit with Dirichlet-inferred probabilities, we derive a tractable approximation of the mutual information between network parameters and latent representations. Our approach not only improves the efficiency of the model but also enhances its robustness under covariate shift. Empirical results on benchmark image classification tasks demonstrate the effectiveness of our method.

1 Introduction

Bayesian neural networks (BNNs) offer a principled way to quantify uncertainty by placing distributions over network parameters. However, traditional weight-based BNNs often face scalability challenges, particularly for large architectures and datasets. To address these issues, Trinh et al. 2022 introduced node-based BNNs, where latent node variables Z are used to simulate implicit input corruptions. In this formulation, variational inference is performed using the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\text{ELBO}}(\theta, \phi) = \mathbb{E}_{q_\phi(Z)} \left[\log p(D | \theta, Z) \right] - \text{KL}(q_\phi(Z) \parallel p(Z)) + \log p(\theta). \quad (1.1)$$

To encourage the model to cover a wider range of corruptions, the authors augment the ELBO with an extra entropy term:

$$\mathcal{L}_\gamma(\theta, \phi) = \mathcal{L}_{\text{ELBO}}(\theta, \phi) + \gamma H[q_\phi(Z)] \quad (1.2)$$

where $H[q_\phi(Z)]$ is the Shannon entropy of the variational posterior and $\gamma \geq 0$ controls the amount of extra entropy.

Whereas this approach gives good experimental results, it has two main theoretical limitations: merely maximizing entropy may lead to have diverse, but uninformative samples, for instance a purely white or black image that would learn nothing. The second limitation is that there is no direct

connection with the output robustness, as the author remark, they only did a proof of concept, not a formal and general link.

Our main contribution is that we show that adding an information term to the loss significantly improves the efficiency of the model.

Our code is fully open source and can be viewed on https://github.com/cvt8/nodes_bayesian.

2 Method

To improve the model, our idea is to add an information term to the loss, i.e.:

$$\mathcal{L}_\gamma(\theta, \phi) = \mathcal{L}_{ELBO}(\theta, \phi) + \gamma H[q_\phi(Z)] - \lambda I(\theta; \phi(G_t)|G_t), \quad (2.1)$$

However, the information computation is intractable. To deal with this, our idea is to consider each vector as a bandit where the probability is inferred using a Dirichlet distribution - as it is the conjugate prior of the multinomial distribution. This idea is also much use to model biological decision processes (Domenech, Rheims, and Koechlin 2020). We could then, using that compute a most precise approximation of the information:

To compute the information we can get from playing a game $n \in \llbracket 1, K \rrbracket$. We denote G_t the action to have played the game at time t . Hence, we want to compute, where the function ϕ is the outcome of the action :

$$I(\theta | \phi(G_t), G_t) = \sum_{k=1}^K \mathbb{E}_t(\theta_k) I(\theta, \phi(G_t) = \theta_k | G_t) = \sum_{k=1}^K \mathbb{E}_t(\theta_k) (H(\theta | G_t) - H(\theta_k | (\phi(G_t) = \theta_k, G_t))) \quad (2.2)$$

We parametrize the Dirichlet distribution with a vector $\alpha = (\alpha_1, \dots, \alpha_K)$ where K corresponds to the size of $\theta = (\theta_1, \dots, \theta_K)$. We can from there compute a close form formula of the information (see Appendix A for the detailed derivation and additional explanations):

$$I(\theta | \phi(G_t), G_t) = \frac{1}{\alpha_0} \sum_{k=1}^K \alpha_k \left[-\ln \alpha_k + \psi(\alpha_k) \right] + \frac{K-1}{\alpha_0} - \psi(\alpha_0) + \ln \alpha_0. \quad (2.3)$$

Where ψ is the digamma function.

2.1 Bandit update rule

Let $\Delta \mathcal{L}_{ELBO}(k)$ denote the change in the ELBO loss when candidate outcome k is chosen. To compute this difference, we need to make two pass in the batch. Experimentally, this doubling trick (Besson and Kaufmann 2018) does not significantly increase compute time, since we not recompute the gradient.

Then, we update the bandit parameters as follows:

- If $\Delta \mathcal{L}_{ELBO}(k) < 0$ (i.e., the outcome improves the loss), update the success parameter:

$$\alpha_k \leftarrow \alpha_k + 1.$$

- Otherwise (if the loss does not improve), we update the other θ_i ($i \neq k$):

$$\forall i \in \llbracket 1, K \rrbracket, i \neq k \Rightarrow \alpha_i \leftarrow \alpha_i + \frac{1}{K-1}$$

In practice, due to the neural network architecture, we don't have access to the chosen θ_k , we therefore need to ponderate them by the softmax probability they are associated to.

Computational resources: We used Constantin's personal laptop GPU Quadro T1000 to make the computations.

3 Results

We conducted experiments over 10 epochs, both with and without our proposed mutual entropy loss. Due to limited computational resources, we aimed to maximize efficiency in training. We observed that beyond 10 epochs, performance improvements became marginal for both models, making additional training less cost-effective. Therefore, we fixed the number of epochs to 10 and maintained identical hyperparameters for both models to ensure a fair comparison.

Figure 1 presents a comparative analysis of the performance of these two models on key benchmarks on the original dataset. Across all evaluated metrics, our model consistently outperforms the vanilla model, demonstrating the effectiveness of our proposed approach.

We evaluate our models using several key metrics:

- **Expected Calibration Error (ECE):** Measures the discrepancy between the predicted confidence and actual accuracy. A lower ECE indicates a better-calibrated model.
- **Negative Log-Likelihood (NLL):** Assesses how well the model assigns high probabilities to correct labels. A lower NLL implies more confident and accurate predictions.
- **NLL_miss:** A variation of NLL that focuses specifically on misclassified samples, providing insights into how the model handles errors.
- **Top-k Accuracy:** The proportion of samples where the correct label is within the top k predicted classes.

One drawback of our method is the increased computational cost during training. This overhead primarily arises from the necessity of running the model twice per batch, compared to a single forward pass in standard machine learning algorithms such as the original implementation. However since the second pass does not require gradient, the extra cost of this second pass is little.

Figure 2 highlights another crucial comparison between the two models, specifically in terms of uncertainty estimation. Our model exhibits a lower aleatoric entropy while maintaining a higher epistemic entropy. This suggests that our model is more confident in its predictions within the data distribution (lower aleatoric uncertainty), while also recognizing its epistemic uncertainty—indicating that it remains aware of regions where it lacks sufficient knowledge. This property is particularly desirable in safety-critical applications, where controlled uncertainty is essential for reliability.

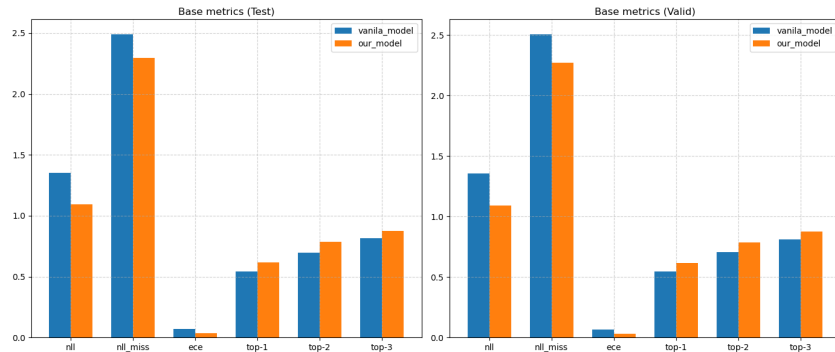


Figure 1: Comparison of the vanilla model and our model across key benchmarks. Our model consistently achieves superior performance.

While our results indicate that our model outperforms the vanilla model across key benchmarks, additional experiments would be necessary to further validate this conclusion. Given more computational resources, training for additional epochs could help ensure that our model is genuinely superior rather than simply learning faster. The observed differences in aleatoric and epistemic entropy suggest that our model is indeed better, as it demonstrates both greater confidence in in-distribution predictions and a higher awareness of epistemic uncertainty.

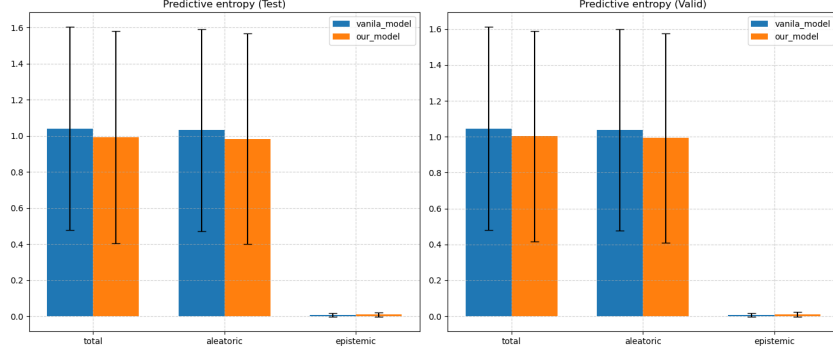


Figure 2: Comparison of aleatoric and epistemic entropy between the two models. Our model exhibits lower aleatoric entropy and higher epistemic entropy, reflecting better calibration and uncertainty awareness.

Moreover, evaluating both models across a broader range of hyperparameters would provide further insights into their relative performance. This would help determine whether our model is intrinsically more effective or if it simply benefits from more favorable hyperparameter settings. Conducting such an analysis would reinforce the robustness of our approach and confirm the generality of the observed improvements.

Corrupted datasets If our model performs better than the original model on the non-corrupted dataset, you may find in Appendix B the plots showing the evolution of different metrics as corruption increases.

The results present a nuanced picture: while our model exhibits a larger performance drop compared to the original model, it consistently achieves higher accuracy. Although Bayesian metrics such as NLL deteriorate more significantly, our model remains the best in terms of classification accuracy, which is ultimately the primary objective.

Thus, while the superiority of our approach is less obvious when considering all metrics, its consistently better accuracy suggests that it still provides a meaningful advantage.

4 Conclusion

In this work, we aimed to improve the handling of covariate shift in node-based Bayesian neural networks by addressing the limitations identified in the original approach. Specifically, we proposed the addition of an information term to the loss function, with the goal of promoting latent representations that are both diverse and informative, while enhancing the model’s robustness. To make this approach feasible, we modeled the problem as a bandit, using Dirichlet distributions to approximate mutual information. The experimental results obtained indicate an improvement in the model’s efficiency compared to the baseline approach, thus supporting the relevance of incorporating information-theoretic principles into variational inference.

This project has enabled us to establish a more explicit connection between information gain and model robustness, while providing a practical implementation, made accessible through the open-source release of our code. These elements provide a foundation for potential future investigations.

Several directions could be considered to extend this work. A first avenue would be to test our approach on other types of neural network architectures, in order to assess its generality. Another possibility would be to explore alternative measures for quantifying information, which could further refine the balance between diversity and relevance in latent representations. Finally, a more in-depth theoretical analysis of the interactions between the entropy and information terms in the loss function could shed light on convergence properties and hyperparameter tuning.

In summary, this project has allowed us to propose a measured improvement to an existing method, drawing on both theoretical and practical tools. It represents an initial step toward a

better understanding of the mechanisms that can enhance the robustness of Bayesian neural networks in the presence of uncertainty.

References

- Besson, Lilian and Emilie Kaufmann (2018). “What doubling tricks can and can’t do for multi-armed bandits”. In: *arXiv preprint arXiv:1803.06971*.
- Domenech, Philippe, Sylvain Rheims, and Etienne Koechlin (2020). “Neural mechanisms resolving exploitation-exploration dilemmas in the medial prefrontal cortex”. In: *Science* 369.6507, eabb0184.
- Trinh, Trung Q et al. (17–23 Jul 2022). “Tackling covariate shift with node-based Bayesian neural networks”. In: *Proceedings of the 39th International Conference on Machine Learning*. Ed. by Kamalika Chaudhuri et al. Vol. 162. Proceedings of Machine Learning Research. PMLR, pp. 21751–21775. URL: <https://proceedings.mlr.press/v162/trinh22a.html>.

A Detailed Dirichlet derivation

A.1 Presentation of the model

In all that follow, we will assume that $\theta_t := (\theta_{1t}, \dots, \theta_{Kt})$, the random variable representing the expected probability $(p_1, \dots, p_K)$ of an outcome at time t , follows a law of Dirichlet of parameter $\alpha_t := (\alpha_{1t}, \dots, \alpha_{Kt})$. Since the Dirichlet distribution of size K , seen as a function of $(x_1, \dots, x_K, \alpha_1, \dots, \alpha_K)$ is an homeomorphism from $[0, 1]^K \times \mathbb{R}_+^*$ to $[0, 1]^K$, any distribution would be equivalent to assign some good parameter α to the initial function. Moreover the dynamical behavior is the good one to model the kind of dynamical Polya urn model we have here.

Moreover, modeling information inferences with Dirichlet processes is experimentally a key to understand the exploration-exploitation dilemma and is what that reports the best to what is happening into the brain (Domenech, Rheims, and Koechlin 2020).

We denote f_t the density function of θ_t .

Remark A.1. For all $n \in \llbracket 1, K \rrbracket$, $\theta_{it} \sim \text{Beta}\left(\alpha_{it}, \sum_{k=1}^K \alpha_{kt} - \alpha_{it}\right)$.

For simplicity, we will assume in the following calculations that $\alpha_0 = (1, \dots, 1)$, i.e. that the initial prior is uniform.

If the prior were not uniform, for example some distribution function $g(x_1, \dots, x_K)$, we would have an initial distribution function of the form $\zeta g * f_0$, where ζ is a normalization factor and $*$ is the convolution product. Thanks to the homeomorphic property of the Dirichlet distribution, it would be equivalent to assigning a good parameter α to the initial function, which would also add information about the elasticity of the information. Thus, all states can be designed by giving an initial condition vector α_0 .

At each time $t \in \mathbb{N}^*$ that a player plays the game, we have the following evolution of the vector α : if the result is $l \in \llbracket 1, K \rrbracket$, then $\alpha_l \leftarrow \alpha_l + 1$.

A.2 The information to be known

Let be $t \in \mathbb{N}$ that is fixed for all the rest of the subsection and can be present only implicitly.

We want to calculate the information we can get from playing a game $n \in \llbracket 1, K \rrbracket$. We denote G_t the action to have played the game at time t . Hence, we want to compute, where the function ϕ is the outcome of the action :

$$I(\theta | \phi(G_t), G_t) = \sum_{k=1}^K \mathbb{E}_t(\theta_k) I(\theta, \phi(G_t) = \theta_k | G_t) = \sum_{k=1}^K \mathbb{E}_t(\theta_k) (H(\theta | G_t) - H(\theta_k | (\phi(G_t) = \theta_k, G_t))) \quad (\text{A.1})$$

We have, for all $k \in \llbracket 1, K \rrbracket$,

$$H(\theta | G_t) = \ln B(\alpha) + (\alpha_0 - K) \psi(\alpha_0) - \sum_{k=1}^K (\alpha_k - 1) \psi(\alpha_k), \quad (\text{A.2})$$

With $\alpha_0 = \sum_{k=1}^K \alpha_k$.

Where B is the multivariate beta function :

$$B(\alpha) = \frac{\prod_{i=1}^K \Gamma(\alpha_i)}{\Gamma(\sum_{i=1}^K \alpha_i)}$$

And ψ is the digamma function :

$$\forall x \in \mathbb{C} - \mathbb{R}_-, \psi(x) := \frac{\Gamma'(x)}{\Gamma(x)}$$

And with,

$$\Gamma(x) := \int_0^\infty t^{x-1} e^{-t} dt.$$

Remark A.2. $\forall z \in \mathbb{C} - \mathbb{R}_-, \Gamma(z+1) = z\Gamma(z)$

Remark A.3. $\forall z \in \mathbb{C} - \mathbb{R}_-, \psi(z+1) = \psi(z) + \frac{1}{z}$.

Remark A.4. We can deduce from the previous remark, with some integral identities that :

$$\forall n \in \mathbb{N} - \{0, 1\}, \psi(n) = -\gamma + \sum_{k=1}^{n-1} \frac{1}{k}.$$

Where γ is the Euler–Mascheroni constant.

In what follows, we may use the notation H_n to talk about the number $\sum_{k=1}^n \frac{1}{k}$.

Hence, the global entropy is the sum of all the local entropy of the variables $(\theta_k)_{k \in \llbracket 1, K \rrbracket}$. With that said, the entropy can be written as :

$$H(\theta|G_t) = \ln B(\alpha) + \left(\sum_{l=1}^K \alpha_l - K \right) \psi \left(\sum_{l=1}^K \alpha_l \right) - \sum_{l=1}^K ((\alpha_l - 1)\psi(\alpha_l)).$$

A.3 Information from a singular option

Let be $k \in \llbracket 1, K \rrbracket$. We denote here δ_k the Kronecker vector : $(0, \dots, 0, 1, 0, \dots, 0)$, where the 1 is at the place k .

$$\begin{aligned} H(\theta_t | (\varphi(G_t) = \theta_k, G_n)) &= \ln B(\alpha + \delta_k) + \left(\sum_{l=1}^K \alpha_l - K + 1 \right) \psi \left(\sum_{l=1}^K \alpha_l + 1 \right) - \sum_{l=1, l \neq k}^K ((\alpha_l - 1)\psi(\alpha_l)) - \alpha_k \psi(\alpha_k + 1) \\ &= \ln B(\alpha) + \ln \frac{\alpha_k}{\sum_{l=1}^K \alpha_l} + \left(\sum_{l=1}^K \alpha_l - K + 1 \right) \left(\psi \left(\sum_{l=1}^K \alpha_l \right) + \frac{1}{\sum_{l=1}^K \alpha_l} \right) \\ &\quad - \sum_{l=1}^K ((\alpha_l - 1)\psi(\alpha_l)) - 1 - \psi(\alpha_k) \\ &= \ln B(\alpha) + \ln \frac{\alpha_k}{\sum_{l=1}^K \alpha_l} + 1 - \frac{K}{\sum_{l=1}^K \alpha_l} + \psi \left(\sum_{l=1}^K \alpha_l \right) + \frac{1}{\sum_{l=1}^K \alpha_l} + \left(\sum_{l=1}^K \alpha_l - K \right) \psi \left(\sum_{l=1}^K \alpha_l \right) \\ &\quad - \sum_{l=1}^K ((\alpha_l - 1)\psi(\alpha_l)) - 1 - \psi(\alpha_k) \\ &= H(\theta|G_t) + \ln \frac{\alpha_k}{\sum_{l=1}^K \alpha_l} + \frac{1-K}{\sum_{l=1}^K \alpha_l} + \psi \left(\sum_{l=1}^K \alpha_l \right) - \psi(\alpha_k) \end{aligned}$$

Hence, we directly have the following result :

$$I(\theta | \varphi(G_t) = \theta_k, G_t) = -\ln \frac{\alpha_k}{\alpha_0} + \frac{K-1}{\alpha_0} - \psi(\alpha_0) + \psi(\alpha_k) \quad (\text{A.3})$$

Or equivalently:

$$I(\theta | \varphi(G_t) = \theta_k, G_t) = -\ln \frac{\alpha_k}{\alpha_0} + \frac{K-1}{\alpha_0} - H_{\alpha_0-1} + H_{\alpha_k-1}. \quad (\text{A.4})$$

A.4 Global information

The global information we can get by playing at the bandit at time t is the pondered sum of all the possible information. We can reclassify, without any loss the (α_k) from the smallest to the largest.

We denote $\alpha_0 := \sum_{l=1}^K \alpha_l$.

Hence :

$$\begin{aligned} I(\theta | \varphi(G_t), G_t) &= \frac{1}{\sum_{l=1}^K \alpha_l} \sum_{k=1}^K \alpha_k \left(-\ln \frac{\alpha_k}{\sum_{l=1}^K \alpha_l} + \frac{K-1}{\sum_{l=1}^K \alpha_l} - \psi \left(\sum_{l=1}^K \alpha_l \right) + \psi(\alpha_k) \right) \\ &= \frac{1}{\alpha_0} \sum_{k=1}^K \alpha_k (-\ln \alpha_k + \psi(\alpha_k)) + \frac{K-1}{\alpha_0} - \psi(\alpha_0) + \ln(\alpha_0) \end{aligned}$$

Hence,

$$I(\theta | \varphi(G_t), G_t) = \frac{1}{\alpha_0} \sum_{k=1}^K \alpha_k \left[-\ln \alpha_k + \psi(\alpha_k) \right] + \frac{K-1}{\alpha_0} - \psi(\alpha_0) + \ln \alpha_0. \quad (\text{A.5})$$

B Metrics on corrupter dataset

In the figures below, n represents the level of dataset corruption, by a gaussian noise, with $n = 0$ indicating no corruption and $n = 4$ corresponding to the highest corruption level.

B.1 Usual metrics

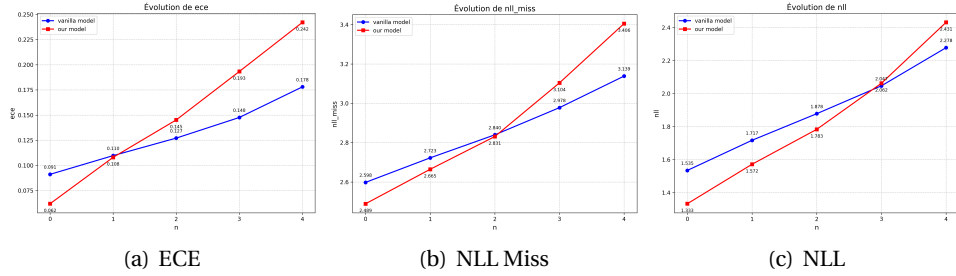


Figure 3: test metrics across different corruption

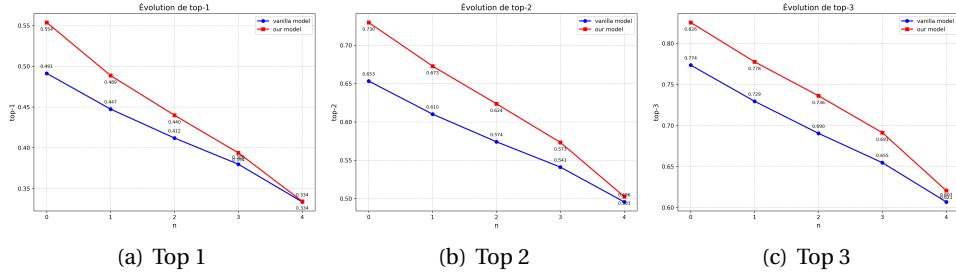


Figure 4: accuracy

B.2 Entropic metrics

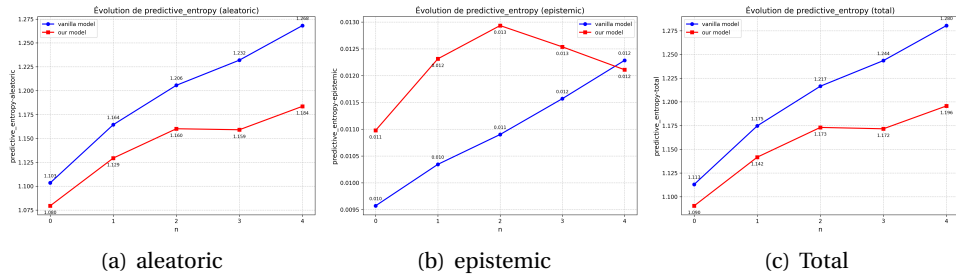


Figure 5: predictive entropy

C Updated algorithmic description

Algorithm 1 ELBO-Based Bandit-Enhanced Variational Inference with Win-Lose Updates

1: Initialization:

- Initialize network parameters θ and variational parameters ϕ for the latent variables Z .
- Choose a prior $p(Z)$ and a variational posterior $q_\phi(Z)$.
- For each pixel (or group of pixels), initialize a bandit model. In the two-outcome case, model each outcome with a Beta distribution with initial parameters α_0 and β_0 . (For K outcomes, use a Dirichlet distribution with parameters $\alpha_1, \dots, \alpha_K$.)
- Define a transformation function φ that maps the input G_t (e.g., pixel values) to a representation used for mutual information estimation.
- Set hyperparameters: entropy weight γ , mutual information weight λ .

2: For each training iteration (over a mini-batch):

1. Forward Pass:

- For each input G_t in the mini-batch, sample $z \sim q_\phi(Z)$.
- Compute the network prediction $f_\theta(G_t, z)$.
- Compute the log-likelihood $\log p(y | G_t, \theta, z)$ for the corresponding label y .

2. Compute ELBO Components:

- Evaluate the expected log-likelihood:

$$\mathcal{L}_{LL} = \mathbb{E}_{q_\phi(Z)} [\log p(y | G_t, \theta, Z)].$$

- Compute the KL divergence $\text{KL}(q_\phi(Z) \parallel p(Z))$.
- Compute the entropy $H[q_\phi(Z)]$.
- Performs a second forward pass for each candidate outcome k without gradient update.

3. Mutual Information Estimation via Bandits: Compute $I(\theta | \varphi(G_t), G_t)$ using equation 2.3.

4. Bandit Reward and Update:

- For each pixel (or group) and candidate outcome k , compute the change in the ELBO loss, $\Delta \mathcal{L}_{ELBO}(k)$.
- If $\Delta \mathcal{L}_{ELBO}(k) < 0$ (i.e., the outcome improves the loss), update the success parameter:

$$\alpha_k \leftarrow \alpha_k + 1.$$

- Otherwise (if the loss does not improve), we update the other θ_i ($i \neq k$):

$$\forall i \in \llbracket 1, K \rrbracket, i \neq k \Rightarrow \alpha_i \leftarrow \alpha_i + \frac{1}{K-1}$$

5. Total Loss Formation:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{ELBO} + \gamma H[q_\phi(Z)] - \lambda I(\theta; \varphi(G_t) | G_t).$$

6. Backward Pass and Update:

- Compute gradients of $\mathcal{L}_{\text{total}}$ with respect to θ , ϕ , and the bandit parameters.
- Update all parameters using an optimizer (e.g., SGD or Adam).

3: Repeat until convergence or for a fixed number of epochs.

4: Inference:

- At test time, sample multiple latent variables $z \sim q_\phi(Z)$ to form a Monte Carlo approximation of the predictive posterior.
 - The bandit parameters, updated via the win-lose rule based on the ELBO improvement, guide the latent representation toward outcomes that improve model performance.
-